Rigid-Body Symmetry in Flow-Matching Motion Planning: What Canonicalisation Buys, and What It Does Not

Internship Report

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Abstract

Motion planning is exactly equivariant under rigid transformations: translate or rotate a planning problem and its solution transforms identically. A neural planner trained on world coordinates does not know this and must relearn the same skill at every pose. This report describes what I built over the internship and what it measured. The work is a *controlled study* of how to supply that symmetry through the problem representation rather than through the network architecture; it is not a proposal for a competitive planner, and I report the classical baselines that make that clear.

I built the benchmark (PointMass3D, with an expert pipeline over RRT-Connect, CHOMP and TrajOpt), then a conditional flow-matching planner over whole trajectories, then the symmetry study. The central observation is that the start and goal of a query themselves determine a frame, and that this frame removes five of the six degrees of freedom of SE(3) exactly and at zero cost. The sixth — a roll about the start–goal axis — cannot be removed by any continuous rule, for two independent reasons, so I handled it by symmetry: roll augmentation during training and frame averaging at inference.

Holding architecture, data and budget fixed and varying only the representation raises the held-out per-sample collision-free rate from 15.4% to 45.6%, a gap of 30.2 points, and the gap *widens* with training data rather than shrinking. The two mechanisms built for the sixth degree of freedom are worth $+0.8 \pm 0.3$ and $+0.1$ points, neither distinguishable from zero at three seed-matched runs. I explain this rather than merely reporting it, by measuring the *non-equivariance residual*: only about 1.7% of the learned velocity field lies outside the equivariant subspace, and frame averaging is exactly the orthogonal projection onto it, so that 1.7% is the whole budget any post-hoc symmetrisation of this model can spend. I am also explicit about what that measurement does *not* license: two arms with indistinguishable residuals differ by 28 points of held-out performance, so the residual does not forecast what a symmetry mechanism is worth. Two further findings support the reading: representation and optimisation are separate bottlenecks, and the whole pattern replicates in a second state space where a state is a full rigid-body pose.

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1 Introduction

A motion planning problem consists of a workspace populated with obstacles, a start configuration $\mathbf{s}$ and a goal configuration $\mathbf{g}$. Its solution is a path τ from $\mathbf{s}$ to $\mathbf{g}$ avoiding the obstacles. The problem possesses an exact symmetry: for any rigid transformation $T \in \mathrm{SE}(3)$, the solution to the transformed problem is the transformed solution,

$$\tau\big(T\cdot\mathcal{O},\, T\mathbf{s},\, T\mathbf{g}\big) \;=\; T\cdot\tau\big(\mathcal{O},\, \mathbf{s},\, \mathbf{g}\big), \tag{1}$$

where $\mathcal{O}$ is the obstacle set. Nothing about the geometry changes when the problem is picked up and moved.

A neural planner trained on raw world coordinates has no access to Eq. (1). As far as its weights are concerned, planning left-to-right near the ceiling is an unrelated task from planning front-to-back near the floor, and the same skill must be learned separately at every pose. Two remedies exist. The first is to build the symmetry into the architecture — constrained layers, irreducible representations, gated nonlinearities — which is laborious to implement, costs expressiveness, and fails in ways that are silent. The second is to build it into the *representation of the problem*, so the network is never shown the pose and therefore cannot be sensitive to it. This report pursues the second, and asks how far it goes.

1.1 What I did, in the order I did it

The internship ran in three stages, and this report follows them.

Stage 1 — the benchmark. I built PointMass3D: a cluttered 3D workspace with an analytic signed distance field, and an expert pipeline combining RRT-Connect, random shortcutting, uniform resampling and CHOMP refinement. Implementing three classical planners rather than one also gave me the *ceiling* the learned planner should be read against (Sec. 2).

Stage 2 — the baseline planner. A conditional flow-matching model over whole trajectories, and the calibration that turned out to matter most: measuring what a straight line and an untrained stochastic prior achieve on the identical problems (Sec. 3, Sec. 6).

Stage 3 — the symmetry study. The $(\mathbf{s}, \mathbf{g})$ reduction, the two obstructions that show the sixth degree of freedom is not removable, the two mechanisms built for it, and the residual diagnostic that explains why they are inert (Sec. 4 onward).

1.2 Contributions

(i) Five of six degrees of freedom are free. The query supplies a frame. Placing the origin at the midpoint of $\mathbf{s}$ and $\mathbf{g}$ and aligning the first axis with the start–goal direction removes three translational and two rotational degrees of freedom exactly, at no computational cost and with no architectural constraint. The resulting representation is invariant under any rigid motion of the problem *up to a single roll about the start–goal axis.*

(ii) The sixth is provably not removable, for two unrelated reasons. A continuous rule fixing the roll would be a nowhere-vanishing tangent field on S^2, which the hairy ball theorem forbids. Deriving the roll from the scene instead fails because the obstacle law is symmetric under central inversion, so the angular moment such a gauge would use vanishes in expectation.

(iii) The symmetry machinery buys little, and I measured why. Rather than infer inertness from flat quality curves — a weak conclusion — I measured the quantity frame averaging exists to remove, and proved exactly what it bounds.

(iv) Calibration, and two bottlenecks. A collision-free rate on a new benchmark means nothing without a scale. Measuring the floor is what revealed that the world-frame arm sits exactly on the straight line, and measuring a longer training run is what revealed that compute and representation relieve different constraints.

I regard (iii) as the transferable part. The residual is a few lines of code, costs one forward pass, needs no retraining, and can be computed on any model with a group action on its inputs. It converts "we tried equivariance and it did not help" into a quantitative statement with an exact bound attached — and it comes with an equally exact statement of its limits, which Sec. 7.4 makes concrete.

2 Stage 1: The PointMass3D benchmark

2.1 Environments and queries

A spherical robot of radius 0.03 moves in the workspace $[-1, 1]^3$, populated with 20 spheres and 20 axis-aligned boxes per environment, with obstacle centres drawn uniformly in $[-0.75, 0.75]^3$. Collision checking uses an analytic signed distance field: the environment SDF is the minimum over all obstacle SDFs and the workspace walls, and clearance subtracts the robot radius. Because the SDF is analytic rather than sampled, clearance is a continuous quantity, which matters later — it gives a graded measure of quality where the binary collision-free rate saturates.

Start and goal are drawn uniformly subject to a collision margin and a minimum separation $\|\mathbf{g} - \mathbf{s}\| \geq 1.5$, so no problem is trivially short. Forty obstacles in a two-unit cube is dense clutter: the direct line between endpoints is blocked on 84% of problems. Absolute success rates are correspondingly low throughout this report and should be read comparatively.

2.2 Expert trajectories

I implemented RRT-Connect, CHOMP and TrajOpt against the same SDF. The expert used for supervision is a pipeline rather than any one of them:

$$\text{RRT-Connect} \rightarrow \text{random shortcutting} \rightarrow \text{uniform resampling to 64 waypoints} \rightarrow \text{CHOMP refinement},$$

with dense collision validation and a fallback to the raw RRT path when refinement fails. Two of those steps deserve a word, because they are not cosmetic.

Shortcutting is the standard post-processing step for sampling-based planners. An RRT-Connect path is a jagged line: the tree grows toward random samples, so the solution wanders and contains detours that have nothing to do with the obstacles. Shortcutting removes them, leaving detours that are actually caused by geometry.

Uniform resampling then fixes a problem shortcutting creates. Shortcutting collapses long free stretches into single segments, so vertex density ends up *inversely* related to geometric complexity — the model would see many waypoints where the path is straight and few where it turns. Resampling to a fixed 64 waypoints fixes both the count and the spacing.

The dataset is 300 environments × 600 start–goal pairs × 30 trajectories, doubled by time-reversal, for 10.8M paths.

2.3 A caveat about dataset size

The dataset is markedly less diverse than its nominal size suggests. The 30 trajectories per start–goal pair are near-duplicates, because CHOMP is a local optimiser and converges to the same route from similar initialisations: the within-pair standard deviation is 0.054 against an overall 0.426. So 10.8M paths carry on the order of 360k distinct queries.

This does not affect any comparison in this report, since both arms train on identical data, but it has two consequences. It dictates the validation split (Sec. 5), and it indicates how additional data should be generated: cutting trajectories per pair from 30 to 5 would buy roughly six times the environments for the same generation cost.

3 Stage 2: The flow-matching planner

The learned planner is a conditional flow-matching model over whole trajectories, in the rectified-flow / conditional-OT formulation. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$. Training draws $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0, 1]$, forms the interpolant $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\,\mathbf{x}_1$ with $\mathbf{x}_1$ an expert trajectory, and regresses the velocity target:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0,\mathbf{x}_1,t}\big\|v_\theta(\mathbf{x}_t, t, c) - (\mathbf{x}_1 - \mathbf{x}_0)\big\|^2, \tag{2}$$

where c encodes the obstacles and the query. Sampling integrates $\dot{\mathbf{x}} = v_\theta(\mathbf{x}, t, c)$ from $t = 0$ to $t = 1$ with explicit Euler steps.

Because the trajectory is a sequence of 64 waypoints, the trunk is a stack of 1D convolutions sliding along the path, with dilations 1–2–4–8 so that the deeper layers can see from one end of the trajectory to the other rather than only between neighbouring waypoints. Conditioning enters through FiLM, fed by a PointNet-style permutation-invariant obstacle encoder. The configuration used throughout has 2.16M parameters.

One architectural detail matters for the cost analysis later: *the conditioning is time-independent*, so the scene is encoded once and reused across all integration steps. That is why frame averaging over K rotations costs $1.05\times$ rather than $K\times$ (Sec. 7.9).

3.1 The Brownian-bridge prior: considered, and not adopted

The flow starts from an isotropic Gaussian that knows nothing about the query, so it must learn both to produce a path and to place its endpoints. A Brownian bridge pinned at start and goal would hand it the endpoints for free, leaving only the obstacle-avoiding detour to be learned.

I did not adopt it, and it is worth recording both the reason I gave at the time and the reason that actually holds, because they are different. My original argument was that a bridge is not isotropic and would break the distributional symmetry of the prior at $t = 0$. That argument is wrong. A bridge with i.i.d. coordinate noise has covariance $\sigma^2\gamma(1-\gamma)I_3$ at path fraction γ — a multiple of the identity — so for any $R \in \mathrm{SO}(3)$ the law satisfies $R \cdot \mathrm{Bridge}(\mathbf{s}, \mathbf{g}) \overset{d}{=} \mathrm{Bridge}(R\mathbf{s}, R\mathbf{g})$, and for a roll about the start–goal axis, which fixes $\mathbf{s}$, $\mathbf{g}$ and every point of the mean segment, the law is invariant outright. The bridge preserves exactly the SO(2) that survives the reduction.

What a bridge prior genuinely costs is two other things: the prior becomes query-dependent, so the guarantee at $t = 0$ becomes an equivariance statement rather than an invariance statement, and the covariance is *singular* at both endpoints, so the regression target there carries no noise. The second is the real hazard and the standard fix is a variance floor. Either way the trade is available, and it is the first thing I would try next.

The bridge did not go to waste. It became the untrained stochastic baseline of Sec. 6.

4 Stage 3: Method

4.1 Decomposing the symmetry group

SE(3) has six degrees of freedom: three translations and three rotations. The key observation is that the query itself supplies a frame, and that this frame consumes five of them.

- **Three translations** are removed by placing the origin at the midpoint $\mathbf{o} = \frac{1}{2}(\mathbf{s} + \mathbf{g})$.
- **Two of the three rotations** are removed by aligning the first axis with the start–goal direction $\hat{\mathbf{x}} = (\mathbf{g} - \mathbf{s})/\|\mathbf{g} - \mathbf{s}\|$. Specifying a direction on the sphere S^2 costs exactly two degrees of freedom, so fixing one axis to it spends two.
- **One rotation remains**: the roll about $\hat{\mathbf{x}}$. Having fixed which way the path runs, nothing in the problem specifies which way is "up" around that axis.

Concretely, let $d = \|\mathbf{g} - \mathbf{s}\|$, let $i^* = \arg\min_i |\hat{\mathbf{x}}_i|$, and set

$$\hat{\mathbf{y}} \propto \hat{\mathbf{x}} \times \mathbf{e}_{i^*}, \qquad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \tag{3}$$

with R the rotation whose rows are $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. A point $\mathbf{p}$ maps to $R(\mathbf{p} - \mathbf{o})$. Because the smallest component of a unit vector satisfies $|\hat{\mathbf{x}}_{i^*}| \leq 1/\sqrt{3}$, the cross product in Eq. (3) has norm $\geq \sqrt{2/3} \approx 0.816$ everywhere, so the construction is well-conditioned over the entire domain with no threshold or fallback branch, and $\det R = +1$ by construction.

What the reduction is invariant to, exactly. This is the point I got wrong first and want to state carefully. Under a rigid transformation T the points $\mathbf{s}$ and $\mathbf{g}$ transform with the problem and the frame transforms with them — but the reduced representation is *not* bit-identical, because the gauge $\hat{\mathbf{y}}$ in Eq. (3) is built from a *fixed world axis* $\mathbf{e}_{i^*}$: rotating the whole problem changes which axis is selected and rolls the frame. What is exactly invariant is everything a roll about $\hat{\mathbf{x}}$ leaves alone — the distance d, the first coordinate of every reduced point, and its radius in the plane normal to that axis — and the entire residual is that one shared roll.

So **the reduction collapses** SE(3) **to** SO(2)**, not to nothing**, and that is exactly why the mechanisms of Sec. 4.3 are needed at all. Within those five degrees of freedom the network never observes the pose and cannot be sensitive to something it is not shown; that part is a stronger guarantee than any trained architecture provides, since it holds to floating-point precision at initialisation rather than to optimisation tolerance. I verify it numerically to 10^{-15} in both domains.

Two consequences follow immediately. The six numbers previously used to condition the network collapse to the single invariant scalar d. And every pair of training examples that were "the same problem in a different pose" now coincide up to a roll, raising the effective density of the training set without generating a single new trajectory.

4.2 Why the sixth degree of freedom is not removable

One might hope to fix the roll by some rule. It cannot be done, and the obstruction is structural rather than a deficiency of any particular construction.

Obstruction from geometry. A rule assigning an "up" direction $\hat{\mathbf{y}}(\hat{\mathbf{x}})$ perpendicular to each axis direction is precisely a continuous, nowhere-vanishing unit tangent vector field on S^2. The hairy ball theorem forbids the existence of such a field. Every deterministic gauge therefore has a discontinuity; one may relocate it but never eliminate it — my i^* tie-break is exactly such a relocated discontinuity, placed on a measure-zero set. The escape route of restricting to a contractible subdomain is closed here: the minimum separation of 1.5 along an axis of extent 2.0 still admits axis-aligned start–goal directions, so the domain is the whole sphere.

For a gauge $\hat{\mathbf{y}}(\hat{\mathbf{x}}, \mathcal{O})$ that also depends on the scene, the bare theorem does not apply — it is a statement about maps out of S^2 alone. Fixing the scene repairs that at the cost of a weaker conclusion: the degenerate set is *per scene*. Every scene admits start–goal directions at which its gauge breaks down, but which directions those are moves as the scene moves, so no restriction of the query distribution avoids them uniformly.

Obstruction from the scene. One might instead derive the roll from the obstacle field, for instance from the angular first moment $c_1 = \sum_j w_j e^{i\varphi_j}$ of the obstacle centres projected into the plane normal to $\hat{\mathbf{x}}$. Let the obstacle-centre law be invariant under the central inversion $\mathbf{p} \mapsto -\mathbf{p}$ and the weights w_j depend only on radius. Inversion fixes every radius and hence every w_j, and maps $\varphi_j \mapsto \varphi_j + \pi$, so $e^{im\varphi_j} \mapsto (-1)^m e^{im\varphi_j}$ and

$$\mathbb{E}[c_m] = 0 \quad \text{for every odd } m \text{ and every } \hat{\mathbf{x}} \in S^2. \tag{4}$$

The restriction to odd m is a fact about the geometry, not a weakness of the proof, and I state it because the natural stronger claim is false — and because I made it. It is tempting to argue that the cube is C_4-symmetric, so only multiples of four survive. That holds when $\hat{\mathbf{x}}$ is a four-fold axis of the cube and fails otherwise, because the plane normal to a general $\hat{\mathbf{x}}$ is not a coordinate plane and the cube's symmetry group does not act on it as C_4. Numerically, at 2×10^5 scenes (noise floor 0.002), $|\mathbb{E}[c_2]|$ is 0.001 along a coordinate axis but 0.033 for a generic direction and 0.070 along a face diagonal, while $|\mathbb{E}[c_1]|$ is at the noise floor for all of them. Central inversion is the hypothesis that actually supports Eq. (4), and it is weaker than uniform-on-a-cube, so the argument covers Gaussian clutter and spherical shells equally.

Eq. (4) is *unconditional*, and a gauge is used *conditionally* — which closes the route more firmly than the statement alone. A scene-derived gauge is computed from the displacements $\mathbf{p}_j - \mathbf{o}$, whose law is centrally symmetric about $-\mathbf{o}$ rather than about the origin, so

$$\mathbb{E}[c_1 \mid \mathbf{s}, \mathbf{g}] \neq 0 \quad \text{whenever} \quad \mathbf{o} \neq \mathbf{0}, \tag{5}$$

with the surviving moment pointing along the projection of $-\mathbf{o}$. I confirm this numerically: at $\mathbf{o} = (0, 0.4, 0.3)$ the conditional first moment has magnitude 0.56 and argument 2.208, against a predicted 2.214. But that direction is not an SE(3)-invariant of the problem — it is a function of where the query sits in the world, and a gauge built on it would reintroduce exactly the world-frame dependence the reduction removes, forfeiting five degrees of freedom to fix the sixth. The first-moment gauge is therefore either degenerate (unconditionally) or not equivariant (conditionally), and neither is usable.

An even moment escapes both, since inversion does not kill even m and an $m = 2$ moment is insensitive to the sign of the displacement. It does not escape the topological obstruction: an $m = 2$ moment defines a *director* rather than a vector, and a continuous director field on S^2 is obstructed for the same Euler-characteristic reason, since $\chi(S^2) = 2 \neq 0$.

Both routes are closed, for unrelated reasons. The sixth degree of freedom must be handled by symmetry rather than canonicalisation.

4.3 Three ways to handle the leftover roll

With canonicalisation ruled out, a symmetry can only be imposed in the data, in the operator, or in the weights. I built the first two and measured them; the third I did not attempt, and Sec. 9 says why it remains open.

Roll augmentation (in the data). During training a uniform angle $\theta \sim \mathcal{U}[0, 2\pi)$ is composed into the same rotation matrix R, so the reduction and the roll are applied as a single transformation rather than two — there is then one place in the code where the point-versus-vector typing of Sec. 4.5 can go wrong, rather than two. The Gaussian prior $\mathbf{x}_0$ is drawn *after* this rotation and never rotated: an isotropic Gaussian is already rotation-invariant, so rotating it would be a redundant step capable only of introducing error.

Frame averaging (in the operator). At inference the velocity field is evaluated at K rolled copies of the state, each result rotated back, and the results averaged:

$$(\mathcal{A}f)(\mathbf{x}) = \frac{1}{K} \sum_{k=0}^{K-1} \rho(\theta_k)^{-1} f\big(\rho(\theta_k)\,\mathbf{x}\big), \qquad \theta_k = \frac{2\pi k}{K}. \tag{6}$$

For $\alpha = 2\pi j/K$ this operator is *exactly* C_K-equivariant regardless of f: rolling the input by α permutes the quadrature points, and reindexing a finite sum is a bijection on them. No architectural constraint and no training assumption is required, and the guarantee holds for an *untrained* network — which is how I test it.

It is worth being precise that the guarantee is exact C_K-equivariance and not SO(2)-equivariance: it holds exactly for rolls that are multiples of $2\pi/K$ and only approximately in between. Drawing a random phase per query trades exact C_K-equivariance for equivariance *in distribution*, which converts systematic aliasing bias into zero-mean scatter.

Frame averaging is not free — the decoder runs K times per integration step — but the expensive half is amortised: because the obstacle encoding is time-independent, the K rolled scene encodings are computed once outside the ODE loop and reused at every step. That is the whole reason the measured cost is 1.05× rather than $K\times$ (Sec. 7.9).

An SO(2)**-equivariant architecture (in the weights).** Not built. Sec. 9 states what that leaves open, and in particular why the smallness of the residual is *not* an argument that it would be low-yield.

4.4 The non-equivariance residual

Frame averaging turns a qualitative question — "is symmetry worth enforcing here?" — into a measurable one, because $\mathcal{A}$ is not merely an averaging operator but a projection. It is linear, idempotent, and self-adjoint with respect to $\langle f, g\rangle = \mathbb{E}_{\mathbf{x}}[f(\mathbf{x})^\top g(\mathbf{x})]$ for any C_K-invariant law on $\mathbf{x}$; hence it is the *orthogonal projection* onto the subspace of C_K-equivariant fields.

Write $v_k = \rho_k^{-1} f(\rho_k \mathbf{x})$ and $\bar{v} = \mathcal{A}f(\mathbf{x})$, and define

$$r = \frac{\left(\mathbb{E}_{\mathbf{x},k}\|v_k - \bar{v}\|^2\right)^{1/2}}{\left(\mathbb{E}_{\mathbf{x}}\|\bar{v}\|^2\right)^{1/2}}. \tag{7}$$

This is the relative norm of the component of the learned field lying outside the equivariant subspace. The norms must be the Hilbert norms of the inner product above, i.e. root-mean-square rather than mean-of-norms; Jensen's inequality separates the two, and only this form makes the bound below an identity rather than an approximation. (An earlier version of my measurement used mean-of-norms in both places, which is one of the corrections in Sec. 8.)

What r bounds. Let f^* be the target velocity field, which is C_K-equivariant because the planning problem is. Since $\mathcal{A}$ is the orthogonal projection onto the equivariant subspace and $\mathcal{A}f^* = f^*$, the error splits orthogonally:

$$\|f - f^*\|^2 = \underbrace{\|\mathcal{A}f - f^*\|^2}_{\text{equivariant error}} + \underbrace{\|(I - \mathcal{A})f\|^2}_{= r^2\|\mathcal{A}f\|^2}. \tag{8}$$

So the squared error that *any* equivariantisation of f can remove is at most $r^2\|\mathcal{A}f\|^2$, and $\mathcal{A}$ attains it; the remaining error is equivariant error, which no symmetrisation of f reduces.

Neither Eq. (8) nor the projection property is novel — both are established in the analysis of equivariant learning — and I claim no novelty for them. What I contribute is the measurement: the size of the anti-equivariant component of a trained planning field, and what it implies for the mechanisms built to remove it.

What r does not bound. Symmetrising a trained field and training a constrained model are different operations. The first is a projection, bounded by Eq. (8); the second changes the hypothesis class, and with it optimisation, capacity allocation and generalisation, none of which r measures. Sec. 7.4 tests the tempting reading directly, and it fails.

4.5 Implementation

Three changes were required to make the reduction representable, and one class of error had to be designed out.

Oriented boxes. The reduction is a general SO(3) rotation, under which an axis-aligned box becomes an oriented box. The original six-number representation (centre and half-extents) cannot express this. I replaced it with a twelve-dimensional representation: a centre and three half-edge *vectors*, which in the world frame are $(h_x, 0, 0)$, $(0, h_y, 0)$, $(0, 0, h_z)$, so the re-encoding is lossless. The environment SDF is unchanged, since collision checking occurs in the world frame after the inverse transform, so no oriented-box distance function is needed. Both arms use this representation, so it is not a confound.

Points and free vectors transform differently. This is the error the implementation is principally designed to prevent, because it fails silently. Waypoints, endpoints, and obstacle centres are *points* and take the full affine map $R(\mathbf{p}-\mathbf{o})$. The three box half-edges are *free vectors* and take the rotation only, $R\mathbf{v}$. Applying the affine map to a half-edge introduces the error $-R\mathbf{o}$ on every edge — identically — so box *position* survives while box *extent* is swamped by a common offset. The model goes blind to obstacle size while the loss curve remains healthy. The severity scales with $\|\mathbf{o}\|$, so the defect is invisible on exactly those problems whose endpoints straddle the origin, which makes it harder still to detect empirically.

Verification. I wrote eighteen tests covering deliberately disjoint failure classes. Asserting that $\mathbf{s}, \mathbf{g}$ land on $(\mp d/2, 0, 0)$ catches translation errors, axis ordering, and R versus $R^\top$, but is blind to handedness, because a reflection in the y–z plane fixes the x-axis pointwise; $\det R > 0$ covers that. Neither detects the point/vector confusion, so a third test asserts featurise$(\rho \cdot S)$ = typed_rotate(featurise$(S), \rho$) with the reference side written as explicit loops rather than the batched operations under test, so that it catches indexing and reshaping errors instead of agreeing with them. A separate suite verifies Eq. (6) against a deliberately untrained network.

One subtlety is worth recording for anyone measuring r. The group acts on the whole input, scene and state together. An early version of my equivariance test rolled only the state while leaving the conditioning attached to its original frame, which produced a large apparent violation from a correct implementation. A diagnostic written that way manufactures the non-equivariance it claims to find.

5 Experimental protocol

All results use environments 250–299 of the 300-environment dataset, which no model was trained on, evaluated as 50 environments × 10 start–goal pairs × 20 samples with 8 Euler steps. The two arms are:

Control world frame, conditioned on the raw pair $(\mathbf{s}, \mathbf{g})$.

Treatment the $(\mathbf{s}, \mathbf{g})$ reduction with uniform roll augmentation, conditioned on the invariant scalar d.

Both arms use the twelve-dimensional oriented-box representation and train on identical trajectories for 20 epochs unless stated otherwise.

Validation split. Within the training environments, validation holds out whole start–goal pairs. A per-trajectory split leaks badly for the reason given in Sec. 2.3: under a per-trajectory split a held-out trajectory has a training neighbour at RMS distance 0.030; under the pair split this rises to 0.162.

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	< 0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	15.4	-0.070	0.067
flow, $(\mathbf{s}, \mathbf{g})$ reduction	**45.6**	-0.010	0.067

Table 1: The same 500 held-out problems, solved five other ways. Classical timings are one CPU core and one query; the learned rows are one GPU and 20 samples per query, so the time column is not a like-for-like comparison. Classical clearance is averaged over solved problems; the learned rows are per-sample rates at 250 training environments.

Two success metrics, kept separate. A sample is *free* if the whole path is collision-free. I report the per-sample rate throughout, because it is the quantity the ablations compare and it is not inflated by drawing more samples. The deployment metric is different: a generative planner is run as best-of-N behind a collision check, so what a user experiences is the fraction of *queries* for which at least one of N samples is free. Mixing the two is the main way this kind of result gets overstated — at $N = 20$ they differ by tens of points.

Statistics. I treat the number of *distinct problems* (500) as the sample size rather than the number of trajectories (10,000), since samples within a problem and problems within an environment are correlated. This gives a standard error of ≈ 2 percentage points, and ≈ 2.7 for a difference of two arms. I checked that assumption rather than asserting it: a bootstrap resampling whole *environments* — the outermost correlated unit — gives 2.2 points on the 15.6% straight-line rate and 2.9 on the 74.0% TrajOpt rate over 10,000 resamples, so the nominal figure is if anything slightly conservative.

Across-seed and across-problem standard errors answer different questions and are not interchangeable. For a claim about a *method* the relevant floor is the across-seed spread, which at 60 environments on the treatment arm is 1.56 points ($n = 3$); ablations sharing a seed with their base are compared paired, where the spread is 0.32. Sec. 8 records what happened when I once compared unpaired.

Loss is not comparable across arms. The treatment regresses targets in the reduced frame, where trajectories are canonicalised along $+\hat{\mathbf{x}}$, so its target variance and hence its mean squared error are mechanically smaller. All reported quantities are world-frame task metrics.

6 Calibration: floors and ceilings

A collision-free rate on a new benchmark means nothing without a scale, so I built one. Table 1 reports four classical planners and a trivial baseline on the identical 500 problems, using the same collision checker, single-threaded on one CPU core.

The floor. A straight line from start to goal is collision-free on 15.6% of these problems. The world-frame model reaches 15.4% per sample. The two are within 0.2 points, an order of magnitude inside the standard error. *On a per-sample basis, whatever the world-frame arm has learned is worth nothing over connecting the endpoints with a line.* This is the sharpest available statement of the problem the reduction solves, and it is invisible without the trivial baseline — which is the main methodological lesson I would carry to the next project.

Prior width σ	0	0.05	0.10	0.30
per-sample free (%)	15.6	11.5	7.4	0.4
best-of-20 (%)	15.6	**15.8**	13.4	2.2
mean path length	1.79	3.31	5.86	16.85

Table 2: The untrained stochastic baseline: Brownian bridges pinned at the endpoints, 20 samples per query, same collision check. Widening the prior buys nothing at any setting, so best-of-N over random detours is not what a learned sampler is competing against.

The stochastic floor. A straight line has one sample, so it cannot be compared against a best-of-N number. The honest control there is the model's own prior with the learning removed: Brownian bridges pinned at start and goal, drawn $N = 20$ times, filtered by the same collision check. I sweep the prior width and report the *best* setting, so the baseline is steel-manned rather than strawmanned (Table 2). Perturbing the straight line buys nothing: the best result over the whole sweep is 15.8% at $\sigma = 0.05$, against 15.6% for the unperturbed line. Random detours leave the workspace faster than they route around anything, and by $\sigma = 0.3$ the mean path is nine times longer than the direct one and solves 2.2%. The $\sigma = 0$ row reproduces the straight-line rate to the digit through an independent code path, which is a check on both harnesses.

This sharpens the reading of the world-frame result in a way worth stating carefully. Per sample, that arm is at the straight-line floor. Per query at $N = 20$ it is not: the world-frame model solves 47.4% of queries with at least one free sample at 250 environments, against 15.8% for the best untrained prior at the same N. So the world-frame model has learned something real — its samples are diverse in an informative way rather than a random one — but it has not learned to make any *individual* sample better than a straight line. The reduction is what closes that second gap.

The ceiling. RRT-Connect solves 99.6% and the expert pipeline 100%, in under 0.4 s of single-core CPU time; CHOMP from a straight-line initialisation solves 88.8%. The learned sampler at 45.6% is not competitive with any of them. I report this plainly because the alternative — comparing a learned planner only against other configurations of itself — is how a benchmark stops measuring anything. The contribution of this work is that the representation change is worth 30 points while the mechanisms built for the residual symmetry are worth at most a fraction of one; that contribution does not require the planner to win, and I do not claim it does.

7 Results

7.1 The reduction scales; the baseline does not

Table 3 and Figure 1 give the principal result. At 250 environments the reduction reaches 45.6% against the baseline's 15.4%, a gap of 30.2 points.

I state the uncertainty on that number carefully, because the obvious phrasing overstates it. The seed spread was measured at 60 environments, not at 250: the seed-matched estimate there is $+20.9 \pm 1.6$ points ($n = 3$ on both arms). The 30.2 at 250 environments rests on one seed per arm, and quoting it as a multiple of a standard error measured in a different cell is not a valid significance statement. Adding seeds at 250 environments is the highest-value missing experiment in this project (Sec. 10).

The trend is more informative than the endpoint. The baseline improves by 2.6 points across a 12.5× increase in training environments: it is close to unable to exploit additional

Training envs	Collision-free (%) World	+aug.	Reduction	best-of-20 Reduction
20	12.9	12.7	19.2	57.8
60	13.6	13.8	34.4	79.6
150	14.6	15.3	41.0	85.6
250	15.4	17.5	**45.6**	**90.4**

Table 3: Held-out performance against training-set size. "+aug." is the world-frame arm trained with random SE(3) motions of the whole problem. The world-frame arm gains 2.5 points across a 12.5× increase in environments and augmentation adds at most 2.1 more, while the reduction gains 26.4 and has not plateaued. The 20- and 60-environment rows are means over three seeds, giving seed-matched differences of $+6.37 \pm 0.19$ and $+20.87 \pm 1.56$; the 150- and 250-environment rows are single seed.

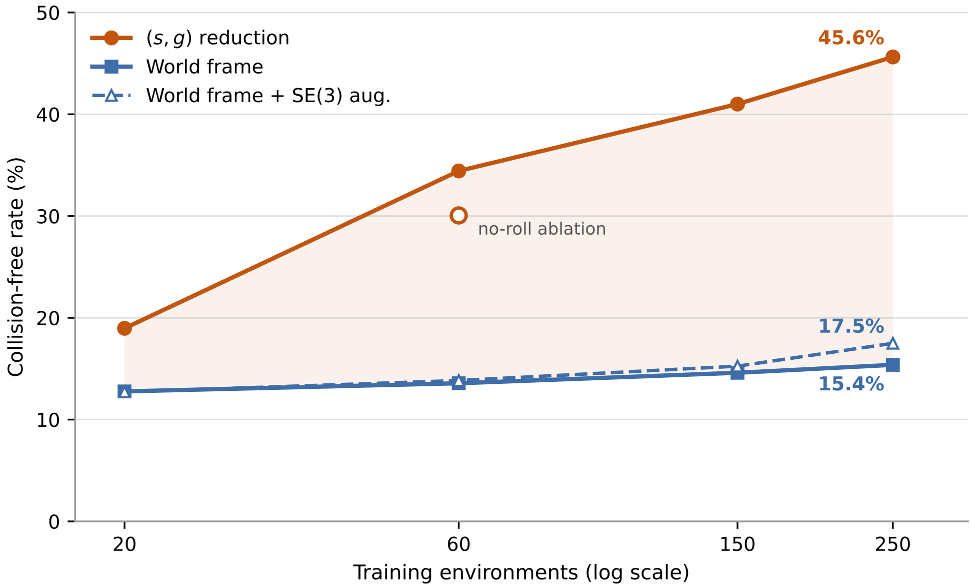

Figure 1: The gap widens with data. If the reduction were compensating for a shortage of training data one would expect it to close; instead the world-frame baseline is nearly flat while the reduced model continues to improve. The hollow marker is the no-roll ablation, drawn on the treatment line because it is the same arm with one component removed.

layouts, because each new layout arrives at poses it must learn about separately. The reduction improves by 26.6 points over the same range and has not levelled off. This is the opposite of what a small-data-crutch account predicts, under which the advantage should shrink as data accumulates. Minimum clearance corroborates through an independent continuous measure: the treatment moves from -0.058 toward -0.0095, approaching feasibility, while the baseline is static.

Augmentation is not the same lever. The comparison against the world-frame arm shows what *ignoring* the symmetry costs. The comparison a practitioner who already knows about the symmetry would want is against the other standard way of using it: augmenting training with random rigid motions of the whole problem. It helps, it helps more as data grows — augmentation needs data to exploit — and it is not close: 12.7, 13.8, 15.3, 17.5 against the reduction's 19.0, 34.4, 41.0, 45.6. At the largest scale augmentation captures about 7% of what canonicalisation does, and leaves the world-frame arm within 2 points of the straight-line floor. Making equivariance *likely* through the data is a different and much weaker mechanism than

Mechanism	Δ free	Assessment
$(\mathbf{s},\mathbf{g})$ reduction (5 DOF)	+30.2 pp	single seed at 250 envs; grows
of which: 3 translations	+12.3 pp	recentring alone
of which: 2 rotations	+18.0 pp	aligning the axis
SE(3) augmentation (world frame)	+2.1 pp	real, but 7% of the above
Roll augmentation	+0.8 pp	$\pm$0.3 paired, n=3; n.s.
Frame averaging (K:1$\rightarrow$3)	+0.1 pp	0.05 SE ($\approx$ 0)

Table 4: Decomposition by mechanism on the same held-out set. The Δ column is against the arm without that mechanism, so the rows are not additive; the two indented rows decompose the row above them.

removing the degrees of freedom outright.

Endpoint error. This falls by a factor of three to five under the reduction ($0.0101 \rightarrow 0.0041$ at 20 environments, $0.0059 \rightarrow 0.0020$ at 250), a direct consequence of canonicalisation: with endpoints always at $(\mp d/2, 0, 0)$ the model no longer spends capacity on *where* they are.

7.2 The mechanisms for the sixth degree of freedom

Roll augmentation. Removing it — the reduction with a fixed gauge — costs 34.4% against 33.5% at 60 environments. Both arms were run at the same three seeds, so the comparison must be paired, and paired it is +1.2, +0.2 and +1.1 points: a mean of $+0.8 \pm 0.3$, consistently positive but $t = 2.6$ on two degrees of freedom, which does not clear significance at $n = 3$. Unpaired, the across-arm spread swamps it entirely.

Frame averaging. On the best model, 45.63% at $K = 1$ and 45.70% at $K = 3$ — 0.05 across-seed standard errors. It bought +0.1 points.

The reduction itself decomposes. A translation-only variant, which recentres on the query midpoint without rotating, reaches 27.7% at 250 environments, between the world frame's 15.4% and the full reduction's 45.6%. So the three translational degrees of freedom are worth +12.3 points and the two rotational ones a further +18.0: 41% and 59% of the total. Neither half carries the effect on its own, which is worth knowing — recentring is the trivially easy part of the reduction and it accounts for two fifths of the benefit.

7.3 Why they are inert: the measured residual

Rather than infer inertness from flat curves, I measured r. On the best model $r = 0.0168$ at $K = 3$: under 2% of the field is non-equivariant. By Eq. (8) the squared error any equivariantisation can remove is at most r^2 of the field's squared magnitude, i.e. under 3×10^{-4}, while the model is failing on more than half of its samples. The two quantities are not the same order, and no rearrangement of the symmetry can close the gap. The remaining error respects the symmetry and is untouchable by symmetrisation.

Two observations sharpen this.

First, removing roll augmentation raises the residual by 64% ($0.0170 \rightarrow 0.0279$ at 60 environments) — a large relative change, but between two values both under 3%, so augmentation is not what makes the field near-equivariant in the first place. The explanation is that the gauge is a deterministic function of $\hat{\mathbf{x}}$, and 600 start–goal pairs per environment supply 600 distinct directions; the reduced-frame training data already spans a wide spread of rolls, and explicit augmentation was solving a problem the data had already solved.

Arm (250 environments)	r under SE(3)	Free (%)
World frame	0.0881	15.4
World frame + SE(3) augmentation	0.0181	17.5
$(\mathbf{s}, \mathbf{g})$ reduction	0.0174	**45.6**

Table 5: The residual does not predict what a mechanism is worth. Augmentation and canonicalisation leave the learned field equally equivariant, and differ by 28 points of held-out performance.

Second, and contrary to what I expected, the residual does *not* move with scale: 0.0183, 0.0170, 0.0177, 0.0168 across a 12.5× range of training data. It is small and flat. This is one of my corrections: an earlier two-point reading under a mean-of-norms definition suggested a downward trend, and I built an argument on that trend which does not survive the four-scale measurement (Sec. 8).

That the residual is flat rather than falling is itself informative. The near-equivariance of the learned field is not something the model acquires gradually from seeing more layouts — it is present at 20 environments and unchanged at 250.

7.4 What the residual does not predict

To test the tempting reading directly I measured r under the *full* SE(3) action rather than the roll alone, which makes it defined for the world-frame arm as well: K random rigid motions are applied to the whole problem, the field is evaluated in each, and each result is mapped back by the inverse rotation.

Augmentation drives the residual down by a factor of 4.9: it demonstrably teaches the network the symmetry, to the same degree the reduction enforces it by construction. It buys 2.1 points. The reduction, at an indistinguishable residual, buys 30.2. *Two models with the same r can differ by* 28 *points*, so r does not forecast the benefit of a symmetry mechanism, and a decision rule of the form "build the constrained architecture only if r is large" has no support here.

One number in Table 5 deserves saying out loud. The world-frame arm ignores the symmetry entirely and performs at the straight-line floor — and it is still 91% equivariant. A model that has learned nothing useful about planning has nonetheless landed close to the equivariant subspace, because most of what a velocity field does on this problem is roughly pose-independent regardless of how it was trained. That is the sharpest form of the point: r being small is compatible with having solved the problem, with having learned the symmetry and nothing else, and with having learned nothing at all.

7.5 Two bottlenecks, not one

Every arm above is trained for 20 epochs, and that budget is not enough. I registered the following prediction in writing before running the experiment, and it came out inside the predicted band.

Holding the data fixed at 60 environments and training the reduced arm for 60 epochs raises the held-out collision-free rate from 34.4% — the mean over three seeds — to **42.4**%, with minimum clearance improving from −0.031 to −0.013. Three times the optimisation, at identical data and identical capacity, is worth +8.0 points: nearly twice the largest of the mechanisms built for the sixth degree of freedom.

The part I did not anticipate is that the same tripling is worth almost nothing to the world-frame arm: 13.6% to 14.4%, an interaction of +7.1 points at roughly 4.5 across-seed standard errors. The two mechanisms are not substitutes; they relieve different constraints. The *representation* bottleneck — the network being shown a pose it must relearn at every value

60 environments	20 epochs	60 epochs	gain
per-sample collision-free (%)			
World frame	13.6	14.4	+0.9
$(\mathbf{s}, \mathbf{g})$ reduction	34.4	**42.4**	+8.0
best-of-20 (%)			
World frame	42.3	39.2	−3.1
$(\mathbf{s}, \mathbf{g})$ reduction	79.6	**87.2**	+7.6

Table 6: Tripling the optimisation at fixed data and fixed capacity. The 20-epoch rows are means over three seeds; the 60-epoch rows are single seed. On the per-sample metric the interaction is +7.1 points; on the deployment metric it is +10.7, and the world-frame arm moves backwards.

Linear probe on the encoder output	World frame	Reduction
scene code, within-environment std	0.000000	0.182
as a fraction of its total std	0.00	0.95
R^2, scene code $\to$ clearance	+0.015	+**0.778**
R^2, full conditioning $\to$ clearance	+0.07	+**0.878**
R^2, separation d alone	—	0.000

Table 7: What the scene code knows about the query. Ridge probes from the 128-d obstacle embedding, and from the full conditioning vector, to the minimum clearance along the straight line from $\mathbf{s}$ to $\mathbf{g}$. Fitted and scored on *disjoint* environments so a probe cannot memorise a scene; the conditioning row is the best score over $\lambda \in [1, 10^4]$.

— is removed by canonicalisation and cannot be removed by compute. The *capacity* bottleneck is removed by compute and cannot be removed by canonicalisation. This also settles the reading of Figure 1: the world-frame arm's flatness is not an artefact of a stingy budget, because giving it three times the budget leaves it flat.

On the deployment metric the effect is stronger and in a direction worth naming. Best-of-20 for the reduction rises from 79.6% to 87.2%, but the world-frame arm *falls*, from a three-seed mean of 42.3% (spread 41.4–43.2) to 39.2%. Tripling the optimisation of the world-frame arm made it worse on the metric that describes how a generative planner is actually deployed. I flag one caveat on my own number: the 60-epoch cells are single seed against three-seed means, which is exactly the shape that produced the error in Sec. 8, so this specific −3.1 needs a seed-matched confirmation before it should be leaned on. The +7.1 interaction on the per-sample metric does not depend on it.

Two points in this domain, but eight-fold in the other. Neither arm here has been trained to a plateau, so the strongest form of the objection — that the gap reflects convergence *rate* rather than representation — is answered in this domain by two points rather than by a limit. The rigid-body domain of Sec. 7.7 tests the same thing over an 8× budget range and reaches the same conclusion, which is why I no longer rest the claim on the 3× comparison alone. A matched convergence study at 60 environments, training both arms until the held-out rate stops moving and scoring snapshots on a fixed epoch grid, is running at the time of writing; Sec. 10 states what each outcome would mean.

7.6 Where the benefit comes from

A 30-point gap invites the question of *why*, and the architecture supplies an answer I can measure. The obstacle encoder max-pools all forty obstacles into a single 128-d vector, and FiLM then applies that vector *identically* to all 64 waypoints.

In the world frame the encoder's inputs do not depend on the query, so its output is constant across every query in an environment. The measured within-environment standard deviation is 0.000000 — exactly zero, because it is forced by the architecture rather than learned. In the reduced frame the obstacles arrive in the query's own frame, so the identical encoder produces a code whose variance is 95% within-environment.

Two things follow, and only the second was obvious to me in advance. First, the max-pool does *not* destroy the query-relevant geometry: in the reduced frame a linear probe recovers how blocked the direct path is at $R^2 = 0.78$ on held-out environments, from that one global vector. Second, the world-frame arm's *full conditioning* — which does contain $(\mathbf{s}, \mathbf{g})$ — cannot express the same quantity linearly across environments, topping out at 0.07 over four orders of magnitude of regularisation. And the treatment's score is not the separation d in disguise: d alone scores 0.000.

So the reduction moves the query–scene interaction out of the network's interior and into the input representation. In the world frame that interaction has to be rebuilt inside the trunk from a per-environment constant plus six query numbers, delivered through a global modulation; in the reduced frame it is already in the features. This is the mechanism behind the finding of Sec. 6 that the world-frame arm never beats the straight line. I state its limits: it is a claim about representation *content*, measured with linear probes on two trained checkpoints, not a causal intervention, and the two arms are different networks, so part of the R^2 gap could reflect learned decodability rather than architecture. The 0.000000 row is immune to that objection, and it is the row the rest depends on.

It also bounds what this encoder can do. A single global code can say *that* a path is blocked; it cannot say *which* waypoint should move and in which direction, since every waypoint receives the same modulation. That is a plausible account of the level at which the reduced arm plateaus — and it is separate from the gap, since both arms share the encoder. I have implemented an oracle diagnostic for it, appending the true SDF and its gradient at each waypoint, which measures the headroom a better encoder could reach without designing one. I report no result from it here: it is the next experiment, not a finding, and it is an upper bound rather than a method, since it supplies exact geometry that no perception-based system would have.

7.7 It replicates in a second state space

The point-mass domain has a convenient property: the state is a point, so the points-versus-vectors distinction of Sec. 4.5 lives only in the obstacle features. To check that the result is not an artefact of that convenience, I built a second domain: an L-shaped rigid body in the same clutter, where a state is a full pose — position plus orientation — rather than a point. The trajectory now carries orientation, so the typing distinction lives inside the state itself. Collision checking evaluates the obstacle SDF at the body's transformed sphere centres; the planner steers by interpolating linearly in position and geodesically (slerp) in rotation, and the trivial baseline is the corresponding geodesic interpolation, which is the SE(3) analogue of joining start to goal with a line.

Three things replicate. The world-frame arm fails to beat the trivial baseline in *both* domains: 1.8 points below the geodesic floor here, 1.2 below the straight-line floor there. The reduction clears its floor in both. And at matched budgets the ratio between the arms is 2.9× in each — an intermediate rigid-body checkpoint reaches 3.0 and 8.6, against 14.4 and 42.4 for the point mass — with the epoch-gain interaction reproducing in sign and shape (+2.1 here against +7.1 there).

This domain also supplies a near-converged test of Sec. 7.5. The cells in Table 8 span an 8× budget range, against the 3× available in the point-mass domain, and the best-validation checkpoints sit at epochs 155 and 159 — so neither arm had stopped improving on

SE(3) rigid body, 250 environments	20 epochs	∼156 epochs	gain
per-sample collision-free (%); trivial floor 4.8			
World frame	2.8	3.1	+0.3
$(\mathbf{s}, \mathbf{g})$ reduction	6.2	**10.0**	+3.8
best-of-20 (%)			
World frame	16.8	15.0	−1.8
$(\mathbf{s}, \mathbf{g})$ reduction	24.8	**40.6**	+15.8

Table 8: The second state space at two budgets an 8× factor apart. The world-frame arm gains 0.3 points across the whole range and never clears its own floor at any budget; the reduction clears it and is still climbing. On the deployment metric the world-frame arm moves backwards, as it does in the point-mass domain. Single seed per cell.

Objective	NFE	World frame	Reduction	Gap
flow matching	8	13.57	34.43	$+20.87 \pm 1.56$
DDPM	8	8.88	23.20	$+14.32 \pm 0.81$
DDPM	100	12.27	28.97	$\mathbf{+16.71 \pm 0.42}$

Table 9: The generative-model control. Same backbone, conditioning, data, frame options and 20-epoch budget at 60 environments; only the objective differs. Three seeds per cell, errors across seeds. Network evaluations are matched within a row.

validation loss when the runs ended. Over that range the world-frame arm gains 0.3 points and the reduction 3.8. The interaction survives at a budget where the reading "both arms are simply undertrained" is much harder to sustain, which makes this the strongest form of the two-bottleneck claim I currently have — stronger than the 3× point-mass comparison a reader is most likely to challenge, and available now rather than pending on the convergence run.

The best-of-20 column carries the sharper version. In the point-mass domain the world-frame arm lost 3.1 points of deployment performance under 3× the budget, and I flagged that as one cell against a three-seed mean. Here it declines monotonically, 16.8 → 15.0, with no seed-matching subtlety, while the reduction rises 24.8 → 40.6. Two state spaces agree that additional optimisation makes the world-frame arm *worse* at best-of-N: it buys a marginal per-sample gain by spending the sample diversity best-of-N actually consumes.

The absolute numbers in the SE(3) domain are small — 10.0% against a 4.8% floor — and I do not want to over-read them. A rigid body in this clutter is a much harder problem than a point, every cell is single seed, and the useful statement is the pattern rather than the level. What the domain establishes is that the reduction is not exploiting something peculiar to a point-shaped state, and that the implementation survives having the typing distinction moved inside the trajectory — which is the failure mode Sec. 4.5 exists to prevent.

7.8 The effect is not specific to flow matching

Every learned number to this point comes from conditional flow matching, which leaves open whether the reduction exploits something about straight-line interpolants rather than something about the problem. So I ran the same two arms under a Diffuser/MPD-style DDPM objective — ϵ-prediction on a cosine schedule — sharing the backbone, conditioning, oriented-box features and frame options unchanged, so the objective is the only difference.

The gap replicates. It is smaller than under flow matching (+16.7 against +20.9, each at its own best integrator budget) and it is nowhere near zero: at three seeds the DDPM difference is over 30 standard errors.

One methodological point is worth recording, because scoring this arm carelessly would have produced the opposite reading. The 8-step operating point used throughout was chosen because straight-line interpolants tolerate it; DDPM does not, and gains roughly four points on *both* arms between 8 and 100 evaluations. Reporting DDPM only at 8 steps would have understated it and invited the fair objection that I had handicapped the baseline with a budget tuned for the method it was meant to test. I report both and read each objective at its own best setting.

The budget caveat applies to both rows equally: these are 20-epoch models, and Sec. 7.5 shows that is well short of convergence, where the flow gap widens from 20.9 to 35.8 points. The DDPM comparison is matched to the flow arm rather than converged, so +16.7 is the value at a common, inadequate budget, not the objective's ceiling.

7.9 Cost

Frame averaging at $K = 9$ costs $1.05\times$ the wall-clock time of $K = 1$ (0.070 vs 0.067 s per query). At 64 waypoints and batch size 1 the sampler is bound by kernel launch overhead rather than arithmetic, and the K rolled scene encodings are computed once outside the ODE loop, so a ninefold increase in width is nearly free. This matters for the interpretation: the finding that frame averaging does not help is *not* a budget trade-off. It was affordable, and I adopted it anyway for the measurement.

Separately, reducing the integrator from 20 to 8 Euler steps costs 0.4–0.8 points and runs $1.4\times$ faster, so 8 steps is used throughout.

8 Corrections

Four claims I made during the internship turned out to be wrong. I record them because in each case the corrected statement is more useful than the original, and because two of them are mistakes that would be easy for someone else to repeat.

1. I thought the reduced representation was bit-identical under rigid motion. It is not. The gauge is built from a fixed world axis, so a global rotation rolls the frame. The correct statement is invariance *up to one roll*, and it is better than the original: if the representation were fully invariant, roll augmentation and frame averaging would have nothing to act on and half of this report would be unmotivated.

2. I reported roll augmentation as worth +4.4 points. It was an artefact of comparing a three-seed mean against a single-seed ablation, where seed 0 happened to be the worst draw for *both* arms (31.3 and 30.1, against 36.0/35.8 and 35.9/34.8). Seed-matched, it is $+0.8 \pm 0.3$ and not significant. The general lesson: the treatment arm's across-seed spread is 1.6 points, five times the paired spread of the ablation, so any ablation compared unpaired at this scale is measuring seed noise rather than the mechanism.

3. I argued the roll gauge could be fixed by a C_4 argument. The claim was that obstacle centres drawn uniformly in a cube give a C_4-symmetric distribution in any coordinate plane, so all moments below $m = 4$ vanish. That holds only when $\hat{\mathbf{x}}$ is a four-fold axis of the cube; for a general direction the normal plane is not a coordinate plane and the cube's symmetry group does not act on it as C_4. The correct hypothesis is central inversion, which kills all *odd* moments for every direction — a weaker conclusion, but one that is actually true, and one that holds for a much broader class of obstacle distributions than my own generator.

4. I reported the residual as falling with scale. That reading came from two points measured under a mean-of-norms definition of r. Under the Hilbert-norm definition of Eq. (7) — the one that makes Eq. (8) an identity rather than an approximation — and measured at four scales, it is flat. I had built an argument on the trend ("the model becomes more equivariant with data, so an equivariant architecture would constrain something already free"), and that argument does not survive. The replacement is Sec. 7.4, which is a stronger result reached by a different route.

9 Discussion and limitations

What the technique contributes. Of the six degrees of freedom of SE(3), five can be removed exactly by a frame read off the query, at no computational cost and with no architectural constraint. Doing so roughly triples the per-sample collision-free rate on this benchmark and, more importantly, converts a model that cannot use additional training environments into one that can. The sixth cannot be removed and, here, does not need to be — a contingent empirical fact rather than a theorem, and one better supported for one mechanism than the other. The K sweep and the residual both bear on *frame averaging*, and are less two independent measurements than two views of one bound, since frame averaging is precisely the projection the residual measures. Neither speaks to roll augmentation, which is a training mechanism rather than a projection, and that is exactly the half the underpowered ablation leaves unresolved.

The learned planner loses to the classical ones. Table 1 is unambiguous: RRT-Connect solves essentially every problem in 0.28 s of single-core CPU time, and the best learned arm solves 45.6% per sample. Part of this is budget — the model is small and briefly trained — and part is that a 64-waypoint sampler from an unconditional prior is simply a weaker instrument than a complete planner on a problem with an exact SDF. I draw the comparison anyway, because a representation result measured only against other configurations of itself is not anchored to anything.

On the underfitting claim, and how it is supported. The obvious support is confounded and I do not use it. Comparing an epoch-averaged training loss against a validation loss computed from exponential-moving-average weights is not a generalisation gap: the training figure lags, since it averages over an epoch during which the weights were still improving, and the two figures come from different weights, with the EMA weights typically the better ones. Both effects push validation below training regardless of fit, so "val < train" cannot be read as evidence of underfitting. I therefore score a fixed slice of the training set with the same EMA weights and the same code path as validation, and read the gap only from that like-for-like pair. The direct evidence is Sec. 7.5: three times the optimisation is worth +8.0 points of held-out collision-free rate, which is what underfitting looks like measured on the task rather than on the loss.

What the residual does and does not bound. Eq. (8) bounds post-hoc symmetrisation of a *given* model, not the achievable performance of a differently parameterised equivariant model. A conclusive test would train an SO(2)-equivariant backbone and compare, and I do *not* use the smallness of r to argue that such an experiment would be low-yield — Sec. 7.4 shows that r does not forecast what a symmetry mechanism is worth, so its smallness cannot be turned around to predict that a constrained architecture would gain little. What rules out *symmetrising this model* is Eq. (8); what a constrained architecture would achieve is untested, and is the largest single gap in this work.

One honest asterisk. Frame averaging *does* consistently improve endpoint error, by roughly 35% ($0.0020 \to 0.0013$), on every checkpoint tested. This is a real effect. It simply does not propagate to the collision-free rate, because endpoint precision was not the failure mode.

Scope of the negative result. The claim that the sixth degree of freedom needs no treatment is specific to this setting, and relies on the training distribution supplying implicit roll diversity through many start–goal directions. A dataset with few directions per environment, or a task whose angular structure is genuinely richer — manipulation with a gripper, or non-holonomic dynamics — could plausibly reverse it. That is precisely why I advocate measuring r rather than generalising my answer: the measurement is cheap and well defined wherever a group acts, even where my conclusion does not carry over.

Seeds only at one scale. The 60-environment cells of both main arms are means over three seeds; every other cell, and every ablation, is a single seed. Even paired, $n = 3$ gives two degrees of freedom and a critical t of 4.3: I can bound the sixth-degree-of-freedom mechanisms as small but cannot resolve whether they are zero.

Seeds are not yet at every scale. The 20- and 60-environment cells of both arms are three-seed means; 150 and 250, and several ablations, remain single seed. That is why any inferential claim in this report is quoted from the seed-matched 60-environment cell.

The generative-objective concern that stood here in an earlier draft — that the effect might be specific to straight-line interpolants — is now measured rather than open. Sec. 7.8 runs the same two arms under DDPM, and the gap survives at $+16.7 \pm 0.4$.

10 What I would do next

In priority order, with the reasoning for the ordering.

1. Seeds at 250 environments, both arms. The headline 30.2 rests on one seed per arm. This is the cheapest experiment on the list and it fixes the most serious defect in the current evidence, so it goes first.

2. The convergence study. Training both arms at 60 environments to a plateau converts Sec. 7.5 from a two-point trend into a limit. If the world-frame arm stays flat while the reduction plateaus higher, "compute cannot buy what the representation forecloses" becomes measured rather than extrapolated. If the world-frame arm climbs, the two-bottleneck framing is wrong and the contribution becomes a claim about convergence rate and sample efficiency, which is weaker but still real — and reporting it after registering the opposite prediction is worth more than the framing it would cost.

3. An SO(2)-equivariant backbone. The comparison this work explicitly cannot make. It is the first question a reader will ask, and the honest answer is currently "untested".

4. The Brownian-bridge prior. For the reasons in Sec. 3.1: the symmetry objection I originally raised does not hold, and the guarantee it was protecting has been measured to be worth 0.1 points. The real cost is the singular endpoint covariance, and the standard fix is a variance floor.

5. More environments, generated differently. The scaling curve has not plateaued at 250 environments, which is the maximum available with 50 held out. Sec. 2.3 indicates how to afford more: cutting trajectories per pair from 30 to 5 buys roughly six times the environments for the same generation cost. I no longer read Figure 1 as saying environments are the single binding constraint, since Sec. 7.5 shows optimisation buys a comparable amount at fixed data; the honest statement is that both are binding and neither has been pushed to saturation.

11 Conclusion

The model presented here is exactly SE(3)-equivariant by construction for five of the six degrees of freedom, and measured to be about 98% equivariant for the sixth without any mechanism enforcing it. The benefit is concentrated in the five: the mechanisms built for the sixth are worth $+0.1$ and $+0.8 \pm 0.3$ points against $+30.2$, and neither is distinguishable from zero at three seeds. Within the five, recentring and axis alignment split the benefit 41/59.

The recommendation is correspondingly simple: canonicalise the degrees of freedom that admit a continuous canonical form — the query usually supplies more of them than one expects — and then *measure* the residual non-equivariance rather than assuming it. The measurement costs one forward pass.

What that measurement licenses is worth stating exactly, since it is narrower than it is tempting to write. It says that *symmetrising this model* is worth almost nothing here, because almost none of this model's error is a symmetry violation. It does not say that a network built equivariant from the start would fail, since such a network searches a different hypothesis class and could converge somewhere better. The claim is about where the error lives, not about what a different architecture could achieve.

The result is a canonicalisation result rather than an equivariance one, and it is more useful for being less elaborate: the requirement is not a specialised architecture but a discipline of not showing the network information it should not have — and then checking, rather than assuming, whether what remains needs treatment.